

Def: $k \geq 1$, M - bip. map. (orient or not) is

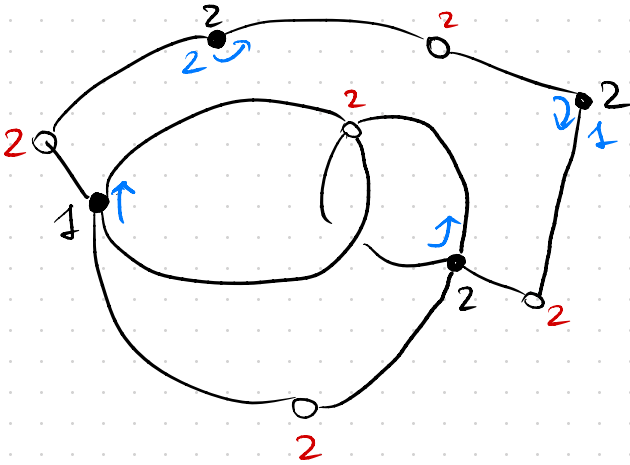
(A) $V_{\bullet}(M) = \bigcup_{i=1}^k V_{\bullet}^{(i)}(M)$ k-layered $V_{\circ}(M) = \bigcup_{i=1}^k V_{\circ}^{(i)}(M)$
 $\forall v_{\circ} \in V_{\circ}^{(i)} \exists v_{\bullet} \in V_{\bullet}^{(i)} v_{\circ} - v_{\bullet}, \forall v_{\bullet} - v_{\circ} v_{\bullet} \in V_{\bullet}^{(j)}(M)$
with $j \leq i$

(B) $V_{\bullet}^{(i)}$ - partition of degrees of $v \in V_{\bullet}^{(i)}(M)$

A k -layered map is labelled if

(A) $\forall j \geq 1 \{v \in V_{\bullet}^{(i)}(M) \mid \deg(v) = j\}$ is
 labelled by $z_1, \dots, z_j(V_{\bullet}^{(i)})$

(B) $\forall v \in V_{\bullet}(M) \exists$ rooted oriented corner



Let

$$F(k) = \sum_{M \in \mathcal{M}(k)} (-t)^{|M|} p_{V_{\Delta}(M)} \frac{b^{\gamma(M)}}{2^{|V_{\circ}(M)| - cc(M)} \alpha^{cc(M)}} \prod_{i=1}^k \frac{(-\alpha s_i)^{|V_{\circ}^{(i)}(M)|}}{2^{V_{\bullet}^{(i)}(M)}}$$

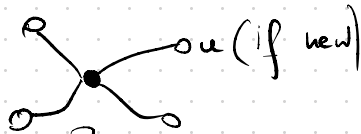
GOAL: $F(k)$ is shifted symmetric.

Lemme: $F^{(k)}(-t; p, s_1, \dots, s_k) = \exp(B_\infty(-\alpha s_1)) \dots \exp(B_\infty(-\alpha s_k)) \cdot 1$,

where $B_\infty(u) := \sum_{n \geq 1} \frac{t^n}{n} B_n(p, u)$;

$$B_n(p, u) = \Theta_Y(\Gamma_Y + u\gamma_+)^n \frac{\gamma_0}{1+b}$$

adds a black "star"



Additionally, $[B_n(p, u), B_m(p, u)] = 0 \quad \forall n, m \geq 1$

Note that:

$$F^{(k)} =^{(*)} \exp(B_\infty(-\alpha s'_1 - 1)) \dots \exp(B_\infty(-\alpha s'_k - k)) \cdot 1,$$

where $s'_i := s_i - i/\alpha$.

Symmetry in $s'_1, s'_2, \dots, s'_k$ is not clear...

Idea: Find more commutation relations to rewrite (*).

$$B_\infty(u) = \sum_{n \geq 1} \frac{t^n}{n} B_n(p, u) = \sum_{n \geq 0} u^n C_n(p, t).$$

Lemme: $[C_n, C_m] = \begin{cases} 0 & \text{if } n \neq 0 \\ (n+1)C_{m+1} & \text{if } n=0 \end{cases}$ for each $m \geq 0$.

Proof: Very technical, extends ideas from our previous paper [Chapuy, Δ, '22].

The main technique: instead of working with C_m , we work with a more refined

version $Y_{m,n} := [u^n] (\Gamma_Y + u \gamma_+)^{m+n} \frac{\gamma_0}{1+b}$

$\leadsto$ We extend algebra by adding 'catalytic' variables $\gamma_0, \gamma_+, \dots$

$\leadsto$ We extend it even further by adding more alphabets X', Z, Z' and we study relations in this richer algebra

$\leadsto$ we project $\rightarrow$ Details in Section 6.

Let $B_\infty^{>0}(u) = \sum_{i \geq 1} C_i \cdot u^i$

Lemma 1 $B_\infty(u+1) = B_\infty(1) + e^{C_0} B_\infty^{>0}(u) e^{-C_0}$

P.f. $B_\infty(u+1) = \sum_{i \geq 0} C_i (u+1)^i = B_\infty(1) + \sum_{i \geq 1} \sum_{k \geq 1} \binom{i}{k} u^k C_i$
 $= B_\infty(1) + \sum_{k \geq 0} u^k \sum_{i \geq 0} \text{ad}_{C_0}^i / i! \cdot C_k = B_\infty(1) + e^{C_0} B_\infty^{>0}(u) e^{-C_0}$ $\square$

$[C_0, C_u] = (u+1)C_{u+1}$

Lemma 2 $\exp(B_\infty(u+1)) = \exp(B_\infty(u)) \exp(B_\infty(1)) \exp(-C_0)$

P.f. $\exp(B_\infty(u+1)) = \exp(C_0 + B_\infty^{>0}(u) + e^{C_0} B_\infty^{>0}(u) e^{-C_0})$

We know that $[\text{ad}_C^i(A_1), \text{ad}_C^j(A_2)] = 0 \quad \forall i \leq i_1 \leq 2, \quad e, u \geq 0$

with $A_1 = B_\infty^{>0}(1), A_2 = B_\infty^{>0}(u), C = C_0,$

$A_1 = B_\infty^{>0}(1), A_2 = e^{C_0} B_\infty^{>0}(u) e^{-C_0}, C = C_0$

For such A_1, A_2, C : $\exp(A_1 + C) \exp(-C) \exp(A_2 + C) = \exp(A_1 + A_2 + C)$ (*)

Therefore: $\exp(B_\infty^{>0}(1) + C_0 + e^{C_0} B_\infty^{>0}(u) e^{-C_0}) =$
 $= \exp(B_\infty^{>0}(1) + C_0) \exp(-C_0) \exp(C_0 + e^{C_0} B_\infty^{>0}(u) e^{-C_0}) =$

$$\begin{aligned}
&= \exp(B_{\infty}^>(1) + C_0) \exp(-C_0) \exp(e^{C_0}(C_0 + B_{\infty}^>(u))e^{-C_0}) \\
&= \exp(B_{\infty}^>(1) + C_0) \exp(-C_0) \exp(e^{C_0} B_{\infty}^>(u) e^{-C_0})
\end{aligned}$$

Using $\exp(-C_0)(e^{C_0} B_{\infty}^>(u) e^{-C_0})^e = B_{\infty}^>(u) \exp(-C_0)$ we deduce that $\exp(B_{\infty}^>(u+1)) = \exp(B_{\infty}^>(1)) \exp(B_{\infty}^>(u)) \exp(-C_0)$ $\square$

P-f of the shifted symmetric property:

We claim that $F^{(k)}(-t, p, s_1, \dots, s_u) = \exp(B_{\infty}^>(k-1)) \dots \exp(B_{\infty}^>(1)) \times \exp(C_0 + B_{\infty}^>(-\alpha s_1' - k) + \dots + B_{\infty}^>(-\alpha s_u' - k)) \cdot 1$

Step 1: Proof by induction $k=1 \rightarrow$ nothing to prove, $k=2$:

Previous lemma gives: $B_{\infty}(u+1) B_{\infty}(v) \stackrel{(*)}{=} B_{\infty}(v+1) B_{\infty}(u)$

$$\begin{aligned}
\text{Therefore } F^{(2)} &= \exp(B_{\infty}^>(-\alpha s_1' - 1)) \exp(B_{\infty}^>(-\alpha s_2' - 2)) \cdot 1 \\
&= \exp(B_{\infty}^>(-\alpha s_2' - 2) + 1) \exp(B_{\infty}^>(-\alpha s_1' - 2)) \cdot 1 \quad \checkmark
\end{aligned}$$

$$\begin{aligned}
\text{Step 2: } F^{(k+1)} &= \exp(B_{\infty}^>(-\alpha s_1' - 1)) \cdot \exp(B_{\infty}^>(k-1)) \dots \exp(B_{\infty}^>(1)) \times \\
&\quad \times \exp(C_0 + B_{\infty}^>(-\alpha s_1' - k) + \dots + B_{\infty}^>(-\alpha s_u' - k - 1)) \cdot 1.
\end{aligned}$$

$$\begin{aligned}
&\stackrel{\text{by using } (*) \text{ } k-1 \text{ times}}{=} \exp(B_{\infty}^>(k)) \dots \exp(B_{\infty}^>(2)) \cdot \exp(B_{\infty}^>(-\alpha s_1' - k)) \times \\
&\quad \times \exp(C_0 + B_{\infty}^>(-\alpha s_2' - k - 1) + \dots + B_{\infty}^>(-\alpha s_u' - k - 1)) \cdot 1 \\
&\quad \exp(B_{\infty}^>(1)) \exp(B_{\infty}^>(-\alpha s_1' - k - 1)) \exp(-C_0) \stackrel{\text{Lemma 2}}{=}
\end{aligned}$$

$$\begin{aligned}
\text{Finally } &\exp(B_{\infty}^>(-\alpha s_1' - k - 1)) \exp(-C_0) \exp(C_0 + B_{\infty}^>(-\alpha s_2' - k - 1) + \dots + B_{\infty}^>(-\alpha s_u' - k - 1)) \\
&= \exp(C_0 + B_{\infty}^>(-\alpha s_1' - k - 1) + \dots + B_{\infty}^>(-\alpha s_u' - k - 1)) \text{ by Lemma 2 } \square.
\end{aligned}$$

We can conclude the main result:

and $(*)$.

$$\theta_{\mu}(\lambda) = (-1)^{|\mu|} \sum_{H \in \mathcal{H}^{\infty}} \frac{b^{\eta(H)}}{2^{|\mathcal{V}_0(H)| - cc} \alpha^{cc}} \prod_{i \geq 1} \frac{(-\alpha \lambda_i)^{|\mathcal{V}_0^{(i)}(H)|}}{z_{\mathcal{V}_0^{(i)}(H)}}$$

P-f:

Step 1:

$$F^{(k+1)}(s_1, \dots, s_k, 0) = F^{(k)}(s_1, \dots, s_k)$$

$F^{(k)}$ - shifted symmetric

Therefore $F^{(\infty)} = \lim F^{(k)}$ exists $\in S^{(a)}[t, p_1, p_2, \dots]$

$$F^{(\infty)} = \sum_{H \in \mathcal{H}^{(\infty)}} (-t)^{|H|} p_{V_\Delta(H)} \frac{b^{q(H)}}{2^{|V_\Delta(H)| - cc} \alpha^{cc}} \prod_{i \geq 1} \frac{(-\alpha s_i)^{|V_\Delta^{(i)}(H)|}}{z_{V_\Delta^{(i)}(H)}}$$

Step 2: $[t^{|\mu|} p_\mu] F^{(\infty)}$ - shifted symmetric

top $[t^{|\mu|} p_\mu] F^{(\infty)} = \frac{\alpha^{|\mu|} z_\mu}{\alpha^{|\mu|} z_\mu} p_\mu(\lambda_1, \dots, \lambda_k)$ - Lecture 3

$[t^{|\mu|} p_\mu] F^{(\infty)}(\lambda_1, \dots, \lambda_k) = 0$ for $|\lambda| < |\mu|$ - Lecture 4

Characterization theorem $\leadsto [t^{|\mu|} p_\mu] F^{(\infty)} = \theta_\mu(\lambda)$